

Mode 1 — Square (centered / opt)

DP16 (8×4) → DP8 (8×4) → DP8 (4×4) → centered add-and-square

Step 1 — DP16 (8×4)

$$\text{DP16 } (8 \times 4) = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP8 (8×4)

$$\text{DP16 } (8 \times 4) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 } (8 \times 4)}$$

Step 3 — DP8 (4×4)

$$\text{DP16 } (8 \times 4) = 2^4 \underbrace{\sum_{i=0}^7 a_{H,i} b_i}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=0}^7 a_{L,i} b_i}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i} b_i}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=8}^{15} a_{L,i} b_i}_{\text{DP8 } (4 \times 4)}$$

Step 4 — Centered add-and-square (subtract 8 from a_L ; each DP8 (4×4) = $\frac{1}{2} [\sum (x+b)^2 - \sum x^2 - \sum b^2 (+512 \text{ for } a_L)]$)

$$\begin{aligned} \text{DP16 } (8 \times 4) &= 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i} + b_i)^2 - \sum_{i=0}^7 a_{H,i}^2 - \sum_{i=0}^7 b_i^2 \right)}_{\text{DP8 } (4 \times 4)} \\ &\quad + \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i} - 8) + b_i)^2 - \sum_{i=0}^7 (a_{L,i} - 8)^2 - \sum_{i=0}^7 (b_i - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\ &\quad + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=8}^{15} (a_{H,i} + b_i)^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15} b_i^2 \right)}_{\text{DP8 } (4 \times 4)} \\ &\quad + \underbrace{\frac{1}{2} \left(\sum_{i=8}^{15} ((a_{L,i} - 8) + b_i)^2 - \sum_{i=8}^{15} (a_{L,i} - 8)^2 - \sum_{i=8}^{15} (b_i - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \end{aligned}$$

Step 5 — All components

$$\begin{aligned} \text{PE} &= 2^4 \sum_{i=0}^7 (a_{H,i} + b_i)^2 + \sum_{i=0}^7 ((a_{L,i} - 8) + b_i)^2 + 2^4 \sum_{i=8}^{15} (a_{H,i} + b_i)^2 + \sum_{i=8}^{15} ((a_{L,i} - 8) + b_i)^2 \\ \alpha &= 2^4 \sum_{i=0}^7 a_{H,i}^2 + \sum_{i=0}^7 (a_{L,i} - 8)^2 + 2^4 \sum_{i=8}^{15} a_{H,i}^2 + \sum_{i=8}^{15} (a_{L,i} - 8)^2 \\ \beta &= 2^4 \sum_{i=0}^7 b_i^2 + \sum_{i=0}^7 (b_i - 8)^2 + 2^4 \sum_{i=8}^{15} b_i^2 + \sum_{i=8}^{15} (b_i - 8)^2 \\ C &= 0 + 512 + 0 + 512 = 1024 \end{aligned}$$